

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: BLURB | Context: EU/French Pharmaceutical Regulatory NLP — BLURB Assessment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's input content is drawn entirely from PubMed abstracts [Q1, Q53, Q54, Q56, Q57, Q58, Q59, Q64, Q70, Q126, Q127], with the pretraining corpus comprising 14 million abstracts [Q126]. The deployment targets French-language SmPCs, PILs, CTD modules, and patent claims — formulaic, template-governed regulatory documents that are categorically distinct from PubMed research prose. BLURB explicitly excludes even clinical notes on the grounds that they 'differ substantially from biomedical literature' [Q43]; this same logic applies a fortiori to regulatory documents, which differ from research abstracts in genre, register, legal function, and language. No French-language source, no SmPC, no PIL, and no EU regulatory submission document appears anywhere in BLURB's data. Web-confirmed: only the small QUAERO EMEA subcorpus (4 documents) provides any French regulatory NLP content [WEB-19, WEB-20], and even there the entity taxonomy is research-biomedical. The 2025 Frontiers SmPC IDMP study confirms empirically that even current LLMs miss ATC codes and excipient dosages in SmPC extraction [WEB-9], evidencing the magnitude of the content gap.

ID	Evidence
Q1	"we compile a comprehensive biomedical NLP benchmark from publicly-available datasets." (p.1)
Q53	"The BioCreative V Chemical-Disease Relation corpus [35] was created for evaluating relation extraction of drug-disease interactions, but is frequently used as a NER corpus for detecting chemical (drug) and disease entities." (p.7)
Q54	"The dataset consists of 1500 PubMed abstracts broken into three even splits for training, development, and test." (p.7)
Q56	"NCBI-Disease. The Natural Center for Biotechnology Information Disease corpus [18] contains 793 PubMed abstracts with 6892 annotated disease mentions linked to 790 distinct disease entities. We use a pre-processed set of train, development, test splits generated by Crichton et al. [14]." (p.8)
Q57	"BC2GM. The Biocreative II Gene Mention corpus [53] consists of sentences from PubMed abstracts with manually labeled gene and alternative gene entities. Following prior work, we focus on the gene entity annotation. In its original form, BC2GM contains 15000 train and 5000 test sentences. We use a pre-processed version of the dataset generated by Crichton et al. [14], which carves out 2500 sentences from the training data for development." (p.8)
Q58	"JNLPBA. The Joint Workshop on Natural Language Processing in Biomedicine and its Applications shared task [27] is a NER corpus on PubMed abstracts. The entity types are chosen for molecular biology applications: protein, DNA, RNA, cell line, and cell type. Some of the entity type distinctions are not very meaningful. For example, a gene mention often refers to both the DNA and gene products such as the RNA and protein. Following prior work that evaluates on this dataset [34], we ignore the type distinction and focus on detecting the entity mentions. We use the same train, development, and test splits as in Crichton et al. [14]." (p.8)
Q59	"EBM PICO. The Evidence-Based Medicine corpus [44] contains PubMed abstracts on clinical trials, where each abstract is annotated with P, I, and O in PICO: Participants (e.g., diabetic patients), Intervention (e.g., insulin), Comparator (e.g., placebo) and Outcome (e.g., blood glucose levels). Comparator (C) labels are omitted as they are standard in clinical trials: placebo for passive control and standard of care for active control. There are 4300, 500, and 200 abstracts in training, development, and test, respectively." (p.8)
Q64	"ChemProt. The Chemical Protein Interaction corpus [31] consists of PubMed abstracts annotated with chemical-protein interactions between chemical and protein entities. There are 23 interactions organized in a hierarchy, with 10 high-level interactions (including NONE). The vast majority of relation instances in ChemProt are within single sentences. Following prior work [8, 34], we only consider sentence-level instances." (p.8)
Q70	"The Genetic Association Database corpus [11] was created semi-automatically using the Genetic Association Archive." (p.9)
Q126	"For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB." (p.12)
Q127	"The original collection contains over 4 billion words; we filter out any abstracts with less than 128 words to reduce noise." (p.12)

Q43	“Clinical notes differ substantially from biomedical literature, to the extent that we observe BERT models pretrained on clinical notes perform poorly on biomedical tasks, similar to the standard BERT.” (p.6)
WEB-19	QUAERO French Medical Corpus — only public French regulatory-document NLP resource, very small EMEA subcorpus (https://quaerofrenchmed.limsi.fr/)
WEB-20	QUAERO HuggingFace dataset documentation — 4 EMEA documents segmented into 15 sub-documents (https://huggingface.co/datasets/DrBenchmark/QUAERO)
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)

Strengths

- Chemical-entity content in BC5-Chemical and ChemProt [Q53, Q64] partially overlaps with active-substance vocabulary that also appears in regulatory text, providing some lexical transfer signal for INN-adjacent entities
- The DDI corpus [Q68] addresses pharmacovigilance-adjacent content ('drug-drug interactions ... particular focus on pharmacovigilance'), which is the closest BLURB content to a regulatory information-extraction concern

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Q68	“DDI. The Drug-Drug Interaction corpus [21] was created to facilitate research on pharmaceutical information extraction, with a particular focus on pharmacovigilance. It contains sentence-level annotation of drug-drug interactions on PubMed abstracts. Note that some prior work [45, 61] discarded 90 training files that the authors” (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires French-language regulatory phrasing, EMA QRD template language [WEB-1, WEB-2], ANSM-jurisdiction conventions, French posology expressions ('deux comprimés par jour'), and EU-specific nomenclature (E-numbers, Ph. Eur. terms). None of this is present in BLURB's English PubMed corpus [Q1, Q126].

IC-2: Not aligned. BLURB content is research-biomedical English prose [Q1]; deployment culture is EU/French regulatory-legal. The content register (formulaic, template-driven) and language are both mismatched.

IC-3: Yes — BLURB inputs are entirely English PubMed abstracts [Q126, Q127] reflecting Anglophone biomedical research conventions. The benchmark itself notes that 'general domain text is substantively different from biomedical text' [Q12]; the same mismatch exists between PubMed abstracts and EU regulatory documents.

IC-4: INSUFFICIENT DOCUMENTATION — would require recruitment of French regulatory affairs specialists to review BLURB samples for relevance to SmPC/PIL/CTD content; not addressed in the paper.

IC-5: Content validity is severely harmed: every BLURB datapoint is in the wrong language, wrong document genre, and wrong register relative to the deployment. Construct-irrelevant variance is essentially total at the content level.

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Q126	"For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB." (p.12)
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)
WEB-2	https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
Q127	"The original collection contains over 4 billion words; we filter out any abstracts with less than 128 words to reduce noise." (p.12)
Q12	"In fact, the majority of general domain text is substantively different from biomedical text, raising the prospect of negative transfer that actually hinders the target performance." (p.2)
Q43	"Clinical notes differ substantially from biomedical literature, to the extent that we observe BERT models pretrained on clinical notes perform poorly on biomedical tasks, similar to the standard BERT." (p.6)

Information Gaps

- Quantification of lexical overlap between BLURB chemical/drug vocabulary and INN/ATC nomenclature for the deployment's drug list

Requires Expert Verification

- Whether the QUAERO EMEA subcorpus is fit-for-purpose as a domain-adaptation seed for the deployment

Remediation

Gap 1: BLURB contains zero French-language and zero regulatory-document content; all inputs are English PubMed abstracts [Q1, Q126].

Recommendation: Construct a French regulatory corpus using the QUAERO EMEA subcorpus [WEB-19, WEB-20] as a seed, supplemented by EMA-published SmPCs/PILs and ANSM-published French regulatory texts. Treat the DART Italian SmPC corpus [WEB-12] as a development-roadmap precedent.

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